

Second shift theorem:

$$\mathcal{L}\{g(t-a)H(t-a)\} = e^{-as}G(s)$$

Example 6.9.3

Find the inverse Laplace transform of

$$F(s) = e^{-2s} \left(\frac{2}{s} + \frac{3}{s-4} \right) + e^{-3s} \frac{1}{(s-1)^3}$$

Solution.

$$G(s) = 2 \frac{1}{s} + 3 \frac{1}{s-4}$$

change form
so it matches
table

$$g_1(t) = 2(1) + 3e^{4t}$$

$$g_1(t-2) = 2 + 3e^{4(t-2)}$$

$$G_2(s) = \frac{1}{(s-1)^3} = \frac{1}{2} \frac{2}{(s-1)^3}$$

$$g_2(t) = \frac{1}{2} e^t t^2$$

$$g_2(t-3) = \frac{1}{2} e^{t-3} (t-3)^2$$

$$G_3(s-1) \Rightarrow G_3(s) = \frac{2}{s^3}$$

$$g_3(t) = t^2$$

$$\therefore f(t) = \underbrace{(2 + 3e^{4(t-2)})}_{g_1(t-2)} H(t-2) + \underbrace{\left(\frac{1}{2} e^{t-3} (t-3)^2\right)}_{g_2(t-3)} H(t-3)$$

Let's tie all this together in a weird example, in which we will need to take inverse LTs using the second shift theorem. Hold on to your hats!

Example 6.9.4

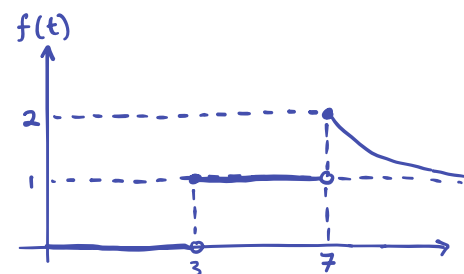
Solve

Second shift theorem:

$$\mathcal{L}\{g(t-a)H(t-a)\} = e^{-as}G(s)$$

when $y(0) = 0$ and $f(t) = H(t-3) + e^{-(t-7)}H(t-7)$.

$$y' + y = f(t)$$



Solution.

1. **TRANSFORM:** $y' + y = 1H(t-3) + e^{-(t-7)}H(t-7)$

$$(sY(s) - y(0)) + Y(s) = e^{-3s} \frac{1}{s} + e^{-7s} \frac{1}{s+1}$$

$$(s+1)Y(s) = e^{-3s} \frac{1}{s(s+1)} + e^{-7s} \frac{1}{(s+1)^2}$$

2.

partial fractions

$$Y(s) = e^{-3s} \left(\frac{1}{s} - \frac{1}{s+1} \right) + e^{-7s} \frac{1}{(s+1)^2}$$

2nd shift:

$$g_3(t) = 1 - e^{-t}$$

$$g_3(t-3) = 1 - e^{-(t-3)}$$

$$g_4(t) = e^{-t}t$$

$$g_4(t-7) = e^{-(t-7)}(t-7)$$

(1st shift thm.) → writing down can be helpful

$$\therefore y(t) = (1 - e^{-(t-3)})H(t-3) + e^{-(t-7)}(t-7)H(t-7)$$

EMTH210 Topic 7: Statistics

Topic Contents

- **Review Material (~2 lectures):** EMTH119 revision/reminder: Random variables and expectation. $E(g(x))$
- **Delta Method (~1 lecture):** Approximating expectations of functions of random variables using Taylor Series.
- **Random Vectors (~2 lectures):** Extending continuous random variables to more dimensions, using multiple integration: Joint distributions, marginal distributions, independence, correlation and a special property of normal random variables. *extending stuff we know*
- **Statistics (~3 lectures):** Sample mean, sample variance, law of large numbers, central limit theorem, sample size and confidence intervals.
- **Hypothesis Testing (~2 lectures):** Making decisions with statistics: Type I and Type II errors, z-test and power. *completely new*
- **Maximum Likelihood Estimation (~2 lectures):** Finding parameter values from data: Likelihood function, maximum likelihood estimator and confidence intervals. λ, μ, σ^2

7.1 Review Material

Probability is the mathematics for quantifying uncertainty. Three key concepts:

1. **Experiment:** An experiment is an activity that results in distinct outcomes that cannot be predicted with certainty. A single performance of an experiment is called a **trial**.
2. **Sample Space:** The sample space Ω is the set of all possible outcomes of an experiment. Subsets of Ω are called events. Consider, for example, the coin toss experiment with two distinct outcomes:

$$\Omega = \{\text{Head}, \text{Tail}\} \quad \text{eg outcomes of a coin}$$

$\approx 24.75\text{mm}$
50c coin width: $\Omega = [24.5, 25]$

3. **Random Variable:** A random variable X is a mapping from the sample space Ω to the set of real numbers

$$X : \Omega \rightarrow \mathbb{R},$$

where each distinct outcome is assigned a unique numerical value.

In this course, we work with random variables directly and rarely mention sample spaces (but they are always there in the background).

Exercise 1 Rat *Symbolic kind of experiment*

A rat is selected at random from a cage of male (M) and female (F) rats. Once selected, the gender of the rat is recorded. Define the sample space and a random variable for this experiment.

real number
from outcome $\Omega = \{M, F\}$

$$X(\omega) = \begin{cases} 1 & \text{if } \omega = M \\ 2 & \text{if } \omega = F \end{cases}$$

one possible random variable

Exercise 2 Tree

A tree is randomly selected from a forest and its height is measured. Define the sample space and a random variable for this experiment.

$$\Omega = (0, \infty)$$

$X = \text{height of tree (in m)}$

eg. for a tree, once it pokes out the soil, it's considered a "tree"

doesn't matter if Ω includes impossible things

Random variables can be either discrete or continuous:

1. **Discrete random variable:** The set of possible outcomes is finite (as in Exercise 1) or at most countably infinite (can be put in a one-to-one correspondence with the integers). *e.g. roll die*
2. **Continuous random variable:** May take on any value in a range (as in Exercise 2) and is formally defined in Section 7.1.1. *e.g. measure die*

separated points (gaps)

interval (no gaps)

We are usually interested in the probability of an event happening and so random variables are often described using distribution functions.

The **distribution function** (or **cumulative distribution function (CDF)**)

$$F_X : \mathbb{R} \rightarrow [0, 1]$$

of the random variable X is

$$F_X(x) = P(X \leq x),$$

Value that changes each time we do an experiment

for any $x \in \mathbb{R}$. That is, the probability that the random variable X takes on a value less than or equal to x .

Theorem: The probability that the random variable X takes a value x in the interval $(a, b]$ is

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

7.1.1 Continuous Random Variables

A random variable, X , is **continuous** if there exists a function f_X such that

1. $f_X(x) \geq 0$ for all x ;
2. $\int_{-\infty}^{\infty} f_X(x) dx = 1$; and
3. for every $a \leq b$,

$$P(a < X < b) = \int_a^b f_X(x) dx.$$

The function f_X is called a **probability density function (PDF)** and the set of values for which f_X is non-zero is called the **support** of X .

$f_X(x)$ is not a probability! Only areas under the $f_X(x)$ curve are probabilities.

The distribution function corresponding to f_X is

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(v) dv.$$

Properties:

1. $0 \leq F_X(x) \leq 1$;
2. $F_X(x)$ is a non-decreasing function of x ; and
3. $f_X(x) = F'_X(x)$ at all points where F_X is differentiable.